

Shapes survey dataset is ready for image classification with clean images but minor table discrepancies.

The shapes_survey.zip dataset achieved an average data quality score of 97.0 out of 100 after cleaning. While the image quality remains high at 99.7 out of 100 with zero corrupt files or duplicate images, the table annotations experienced quality issues including missing image files, unreferenced files, and label disagreements. The cleaned dataset contains 38 total images with a class distribution of 16 circles, 12 squares, and 10 triangles.

Data quality before	After cleaning	Items	Numbers fact-checked
92.4/100	97.0/100	76 → 75	10

Cross-type findings

WARNING Table rows point to missing image files

3 table rows name image files that are missing from the dataset out of 38 total rows.

Recommendation: Locate the missing image files or drop the corresponding rows before model training.

Evidence: mix_links_001

WARNING Dataset contains unreferenced image files

4 image files out of 38 total files have no corresponding annotation row in the table.

Recommendation: Annotate the unreferenced images or exclude them from training.

Evidence: mix_links_001

CRITICAL Label disagreements between table and directory folders

There are 3 label disagreements where table labels conflict with image directory paths, despite 35 rows matched.

Recommendation: Manually review the conflicting cases to resolve label ground truth prior to training.

Evidence: mix_links_001

INFO Table quality issues impact dataset reliability

Table quality issues including row link errors and label conflicts reduced table quality from 85.1 to 94.3 out of 100, which remains lower than image quality at 99.7 out of 100.

Recommendation: Address table labeling and row link issues to bring overall dataset quality up to image quality standards.

Evidence: mix_overview_001, mix_links_001

WARNING Image file referenced by multiple rows

1 image file is referenced by more than one row in the table annotations.

Recommendation: Inspect and clean duplicate row references to ensure each image maps to a single annotation.

Evidence: mix_links_001

Each data type

Each type was profiled, cleaned, analyzed, and reviewed by its own crew. Their full reports are in `cleaned_by_type.zip`.

Table · 1 file(s) · quality 85.1 → 94.3 · rows 38 → 37

- Exact duplicate rows present in raw data: The raw table contained 1 exact duplicate rows, representing 2.63% of the dataset.
- Data quality score improved after cleaning: The overall data quality score increased from 85.10 out of 100 in the raw dataset to 94.30 out of 100 in the cleaned dataset.
- High missingness in notes column: The notes column had 44.74% missing values in the raw data, which was handled during cleaning.
- Confidence score distribution characteristics: The confidence score has a mean of 85.89 and a median of 86.00 in the raw dataset.

Image · 38 file(s) · quality 99.7 → 99.7 · images 38 → 38

- Small total dataset size: The cleaned dataset contains 38 total images and 38 analyzed images across all categories.
- Class distribution imbalance: There are 16 circles, 12 squares, and 10 triangles, resulting in an imbalance ratio of 1.6.
- Zero corrupt or low quality images: Quality analysis found 0 blurry images and 0 corrupt files out of 38 analyzed images.
- No duplicate images detected: There are 0 exact duplicate groups and 0 cross-class duplicate files in the dataset.
- High dataset quality score: The cleaned dataset achieves a quality score of 99.7 out of 100 based on evaluated metrics.
- No suspected mislabels in vision review: Based on an AI vision review, there are 0 suspected mislabeled images across all classes.
- Dataset size unchanged after cleaning: Cleaning steps modified 0 images, keeping the total count at 38 based on the overview.

Table-to-file links

annotations.csv, column 'image': 35 of 38 rows match a file; 3 point to missing files; 4 of 38 files have no row; column 'label' disagrees with the folder for 3 rows.

Problem	Example
Row names a missing file	circle_100.jpg
Row names a missing file	circle_101.jpg
Row names a missing file	circle_102.jpg
File with no row	shapes_survey/images/circles/circle_003.jpg
File with no row	shapes_survey/images/squares/square_012.jpg
File with no row	shapes_survey/images/triangles/triangle_005.jpg
File with no row	shapes_survey/images/triangles/triangle_010.jpg
Label disagrees (row 4)	shapes_survey/images/circles/circle_004.jpg: table 'square', folder 'circles'
Label disagrees (row 14)	shapes_survey/images/circles/circle_014.jpg: table 'triangle', folder 'circles'
Label disagrees (row 24)	shapes_survey/images/squares/square_008.jpg: table 'circle', folder 'squares'

Next steps

1. Resolve the 3 label disagreements between table labels and image directory paths.
2. Address the 3 missing image files and 4 unreferenced image files to align table annotations with the file system.
3. Account for the class imbalance across circles, squares, and triangles during model training.

4. Verify the single image file referenced by multiple annotation rows.

How the agents checked themselves

Model calls 19 · guardrail catches 3 · self-corrections 3 · review rounds 4 · model switches 8

+7s Switched models for the strategist

gemini-3.8-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+14s Switched models for the strategist

gemini-3.7-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+15s Switched models for the strategist

gemini-3.7-flash hit its server error 503 (set aside for 120s); trying the next model in the pool.

+17s Switched models for the strategist

gemini-3.6-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+22s Switched models for the strategist

gemini-3.5-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

+30s Findings failed the fact check

Finding 4 ('Class balance and label distribution'): 'label.top_count' isn't in ['tbl_columns_001']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them.

+31s Findings failed the fact check

Finding 3 ('Zero corrupt or low quality images'): the statement uses 38, which isn't in the cited evidence. It matches 'images' in img_overview_001 or 'analyzed' in img_overview_001 or 'formats.JPEG' in img_overview_001: add that key and cite that artifact. Finding 7 ('Dataset size unchanged afte...

+33s Findings failed the fact check

Finding 4 ('Label distribution shows shape frequency'): 'label.top_count' isn't in ['tbl_columns_001']. Use '<column>.<stat>' keys that appear in the evidence, and cite the artifact that contains them.

+35s Revised output accepted

+35s Revised output accepted

+37s Reviewer approved the findings

+37s Reviewer approved the findings

+47s Switched models for the analyst

gemini-3.8-flash hit its server error 503 (set aside for 120s); trying the next model in the pool.

+70s Switched models for the reviewer

gemini-3.6-flash hit its server error 503 (set aside for 30s); trying the next model in the pool.

... and 3 more in the HTML report.

Cleaning applied

Each data type was cleaned by its own crew.

Charts

Data quality by type

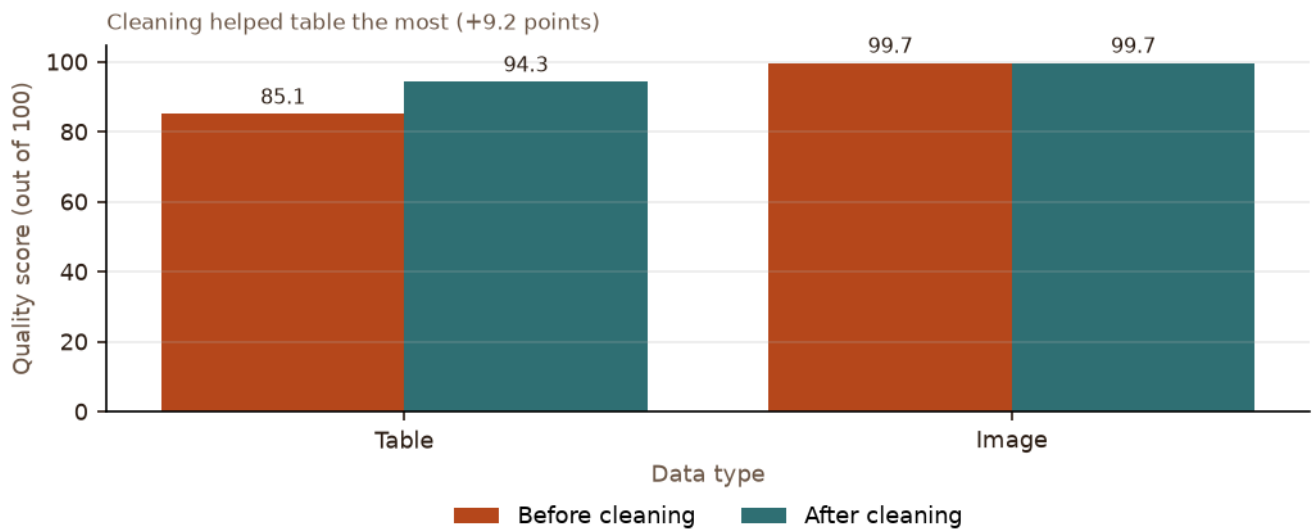


Table: Missing values by column

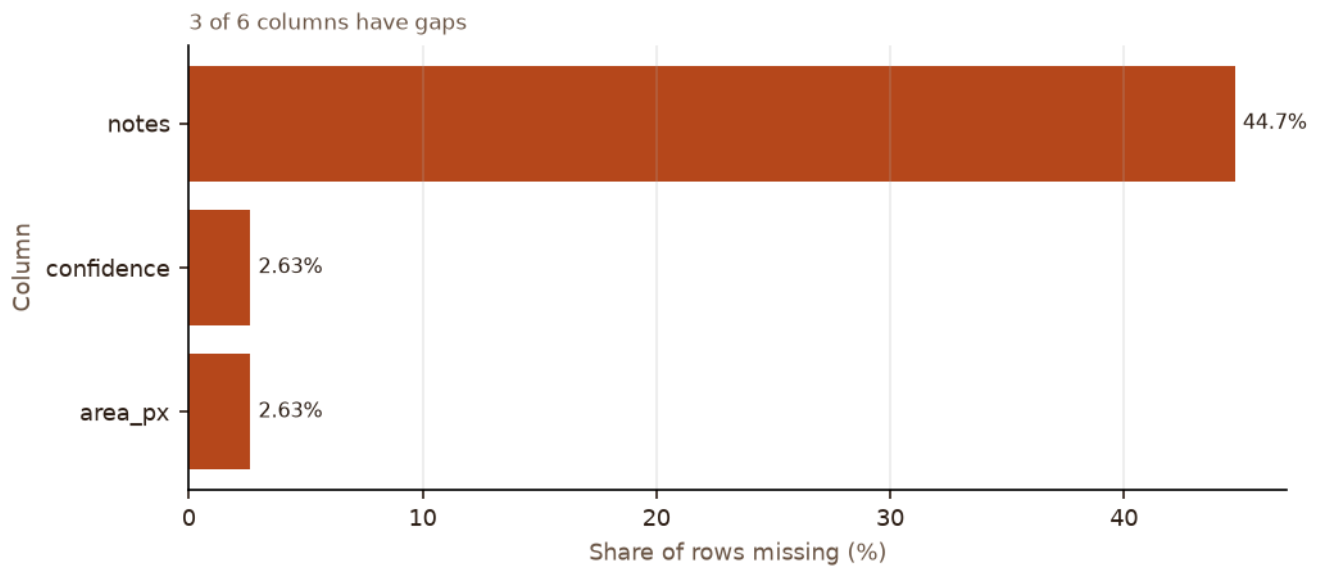


Table: Distribution of confidence

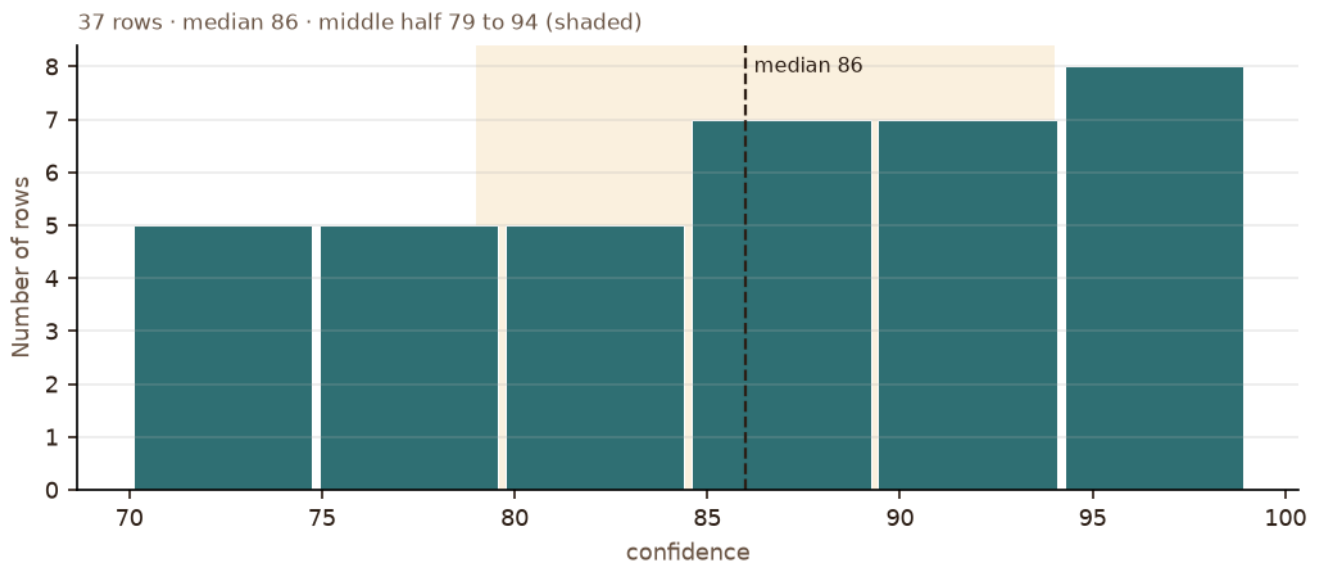


Image: Images per class

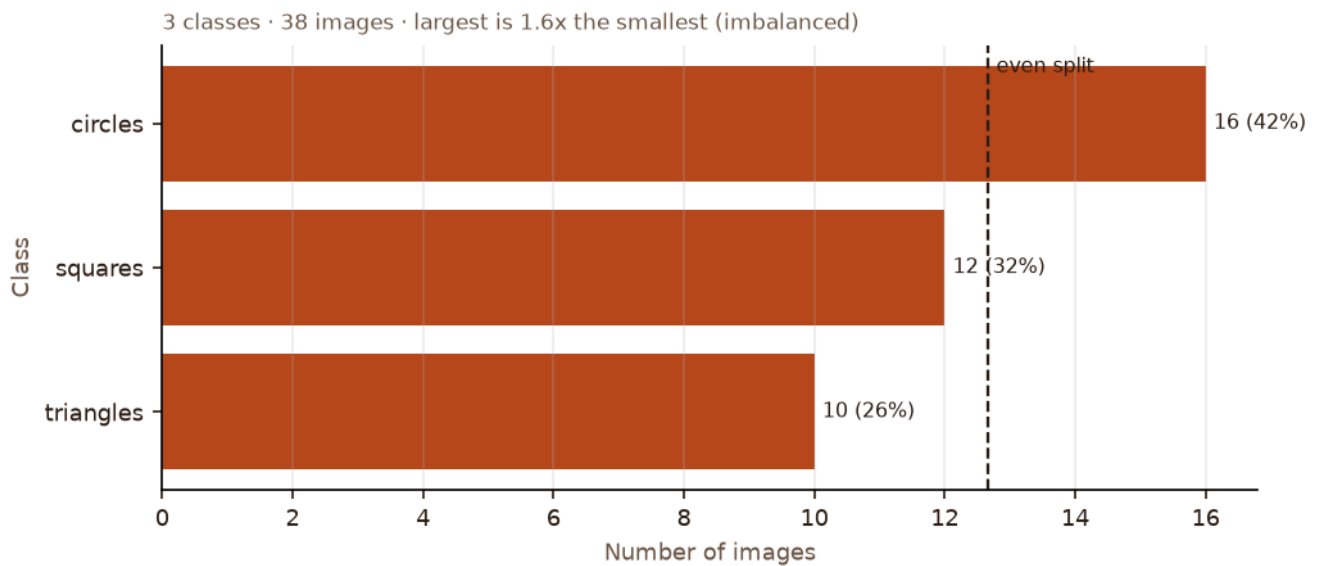
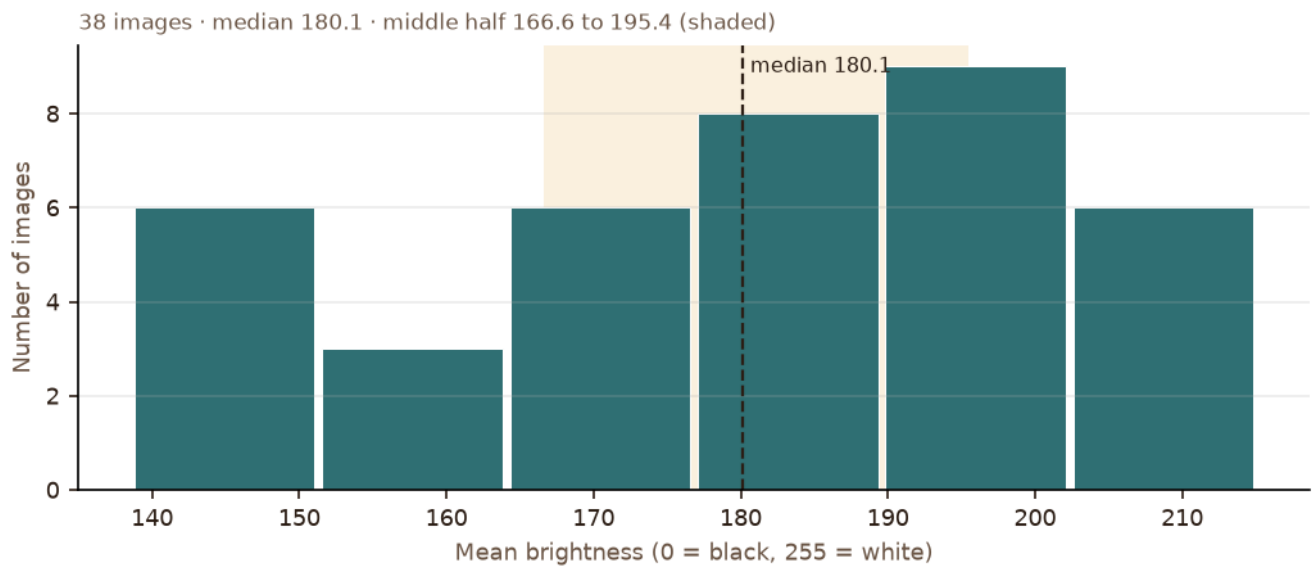


Image: Brightness



Every number in the findings was checked against the evidence artifacts listed. Models used: gemini-3.5-flash-lite, gemini-3.6-flash.